

## BOGOR CITEKO, Indonesia

WMO: 967510

Lat: 6.6980S

Lon: 106.9350E

Elev: 3117

StdP: 13.11

Time Zone: 7.00 (E07)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | 62.4       | 63.2 | 53.9                          | 69.2 | 68.7 | 56.1 | 75.3 | 67.4 | 12.2                  | 71.3 | 9.8 | 71.1 | 0.6                      | N/A  | 0.175 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 11.5                   | 81.8            | 69.6 | 80.7 | 69.9 | 79.9 | 69.9 | 73.1                | 77.8 | 72.6 | 77.4 | 72.1 | 76.8 | 4.6                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 71.6                            | 131.3 | 76.2 | 71.0 | 128.5 | 75.6 | 70.6 | 126.6 | 75.3 | 39.1          | 78.2 | 38.5 | 77.7 | 38.1 | 77.3 | 79.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 9.1               | 7.5  | 6.4 | 58.6                       | 84.8 | 3.4                | 1.4 | 56.1                                               | 85.8 | 54.1       | 86.7 | 52.2       | 87.5 | 49.7       | 88.5 |  |  |
|                   |      |     | 53.5                       | 76.2 | 1.6                | 1.5 | 52.4                                               | 77.2 | 51.5       | 78.1 | 50.6       | 78.9 | 49.4       | 80.0 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 71.5   | 70.0  | 70.2  | 71.5  | 72.4  | 72.5  | 72.1  | 71.1  | 71.2  | 71.7  | 72.1  | 71.7  | 71.3  |
|                                                                        | DBStd       | 1.65   | 2.06  | 1.92  | 1.41  | 1.32  | 1.35  | 1.31  | 1.18  | 1.13  | 1.17  | 1.37  | 1.50  | 1.50  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD50       | 7844   | 619   | 565   | 666   | 672   | 698   | 664   | 654   | 657   | 652   | 685   | 652   | 660   |
|                                                                        | CDD65       | 2369   | 154   | 145   | 201   | 222   | 233   | 214   | 189   | 192   | 202   | 220   | 202   | 195   |
|                                                                        | CDH74       | 6117   | 174   | 169   | 360   | 578   | 712   | 657   | 555   | 638   | 733   | 733   | 467   | 341   |
|                                                                        | CDH80       | 168    | 3     | 4     | 3     | 13    | 18    | 11    | 8     | 13    | 31    | 45    | 16    | 3     |
| (14) Wind                                                              | WSAvg       | 1.5    | 1.8   | 1.5   | 1.5   | 1.4   | 1.3   | 1.3   | 1.5   | 1.6   | 1.5   | 1.4   | 1.3   | 1.5   |
| (15) Precipitation                                                     | PrecAvg     | 147.2  | 16.6  | 17.0  | 14.4  | 14.7  | 12.4  | 7.7   | 7.0   | 7.2   | 7.9   | 12.6  | 15.7  | 14.0  |
|                                                                        | PrecMax     | 179.6  | 37.1  | 28.6  | 27.3  | 20.7  | 21.5  | 17.4  | 15.6  | 16.7  | 19.4  | 23.1  | 25.0  | 22.0  |
|                                                                        | PrecMin     | 109.1  | 4.6   | 7.1   | 6.1   | 4.7   | 5.4   | 2.0   | 1.3   | 1.7   | 2.0   | 4.1   | 8.0   | 3.8   |
|                                                                        | PrecStd     | 18.7   | 6.8   | 4.6   | 4.8   | 3.6   | 3.5   | 3.5   | 3.9   | 4.1   | 4.3   | 4.7   | 4.0   | 4.5   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 80.3  | 80.2  | 80.4  | 81.6  | 82.0  | 81.4  | 81.2  | 81.6  | 82.9  | 83.6  | 82.3  | 80.2  |
|                                                                        |             | MCWB   | 70.8  | 70.0  | 71.4  | 71.5  | 70.8  | 70.6  | 68.9  | 68.2  | 67.9  | 67.1  | 69.9  | 70.7  |
|                                                                        | 2%          | DB     | 78.0  | 77.4  | 78.9  | 80.1  | 80.2  | 80.0  | 79.4  | 79.9  | 80.8  | 81.1  | 80.0  | 78.7  |
|                                                                        |             | MCWB   | 70.3  | 70.3  | 70.8  | 71.5  | 71.2  | 70.4  | 68.8  | 68.5  | 68.5  | 68.7  | 70.4  | 70.5  |
|                                                                        | 5%          | DB     | 75.8  | 76.1  | 77.5  | 78.9  | 79.0  | 79.0  | 78.3  | 78.7  | 79.5  | 79.6  | 78.2  | 77.4  |
|                                                                        |             | MCWB   | 69.8  | 70.1  | 70.6  | 71.3  | 71.3  | 70.6  | 68.9  | 68.4  | 68.6  | 69.4  | 70.6  | 70.3  |
|                                                                        | 10%         | DB     | 74.1  | 74.6  | 76.2  | 77.4  | 77.9  | 77.2  | 77.5  | 77.5  | 78.3  | 78.1  | 77.0  | 76.0  |
|                                                                        |             | MCWB   | 69.3  | 69.8  | 70.4  | 71.1  | 71.2  | 70.5  | 68.9  | 68.6  | 68.7  | 69.7  | 70.6  | 70.2  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 72.4  | 72.5  | 73.2  | 73.7  | 73.8  | 73.2  | 72.6  | 72.0  | 72.0  | 72.5  | 72.9  | 72.8  |
|                                                                        |             | MCDB   | 76.7  | 77.0  | 77.5  | 78.5  | 78.5  | 78.4  | 77.4  | 76.9  | 77.0  | 77.6  | 77.4  | 77.1  |
|                                                                        | 2%          | WB     | 71.4  | 71.6  | 72.2  | 72.8  | 72.9  | 72.4  | 71.4  | 71.0  | 71.1  | 71.7  | 72.0  | 71.9  |
|                                                                        |             | MCDB   | 75.8  | 75.7  | 76.6  | 77.6  | 77.6  | 77.6  | 76.4  | 76.3  | 76.7  | 77.0  | 76.7  | 76.3  |
|                                                                        | 5%          | WB     | 70.5  | 70.9  | 71.5  | 72.1  | 72.4  | 71.8  | 70.6  | 70.3  | 70.6  | 71.1  | 71.5  | 71.2  |
|                                                                        |             | MCDB   | 74.5  | 74.7  | 75.9  | 76.9  | 77.1  | 76.9  | 75.8  | 75.9  | 76.4  | 76.4  | 76.2  | 75.5  |
|                                                                        | 10%         | WB     | 69.6  | 70.1  | 70.8  | 71.5  | 71.8  | 71.1  | 69.9  | 69.7  | 69.9  | 70.5  | 70.8  | 70.6  |
|                                                                        |             | MCDB   | 73.1  | 73.6  | 75.0  | 76.3  | 76.5  | 76.2  | 75.2  | 75.5  | 76.0  | 75.9  | 75.5  | 74.8  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 7.8   | 8.1   | 10.2  | 11.0  | 11.5  | 11.8  | 13.0  | 13.9  | 14.0  | 12.8  | 11.1  | 9.9   |
|                                                                        |             | MCDBR  | 11.3  | 11.2  | 12.2  | 13.0  | 13.3  | 13.5  | 14.7  | 15.3  | 16.1  | 15.3  | 13.0  | 12.1  |
|                                                                        |             | MCWBR  | 6.8   | 7.3   | 7.5   | 7.8   | 7.9   | 8.3   | 8.7   | 8.6   | 8.6   | 8.2   | 7.6   | 7.3   |
|                                                                        | 5% WB       | MCDBR  | 9.9   | 10.0  | 10.9  | 11.4  | 11.5  | 11.8  | 12.3  | 13.0  | 13.1  | 12.3  | 11.4  | 10.7  |
|                                                                        |             | MCWBR  | 6.7   | 6.9   | 7.3   | 7.5   | 7.7   | 7.9   | 8.0   | 8.1   | 7.9   | 7.8   | 7.4   | 7.1   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.426 | 0.451 | 0.461 | 0.489 | 0.494 | 0.486 | 0.494 | 0.518 | 0.538 | 0.584 | 0.572 | 0.472 |
|                                                                        | taud        |        | 2.347 | 2.272 | 2.241 | 2.148 | 2.112 | 2.111 | 2.074 | 1.985 | 1.938 | 1.831 | 1.880 | 2.203 |
|                                                                        | Ebn at Noon |        | 290   | 282   | 275   | 259   | 248   | 245   | 246   | 247   | 251   | 244   | 248   | 275   |
|                                                                        | Edh at Noon |        | 42    | 46    | 46    | 49    | 48    | 47    | 50    | 57    | 62    | 70    | 67    | 48    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1460  | 1420  | 1555  | 1552  | 1525  | 1477  | 1554  | 1724  | 1802  | 1684  | 1526  | 1481  |
|                                                                        | RadStd      |        | 142   | 132   | 95    | 77    | 78    | 107   | 124   | 88    | 138   | 155   | 94    | 108   |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (6 neighbors) | +0.86 | +1.10   | N/A | +0.76  | N/A     | N/A   | N/A   | N/A         | N/A   | +238  | +238  |

Nomenclature: See separate page